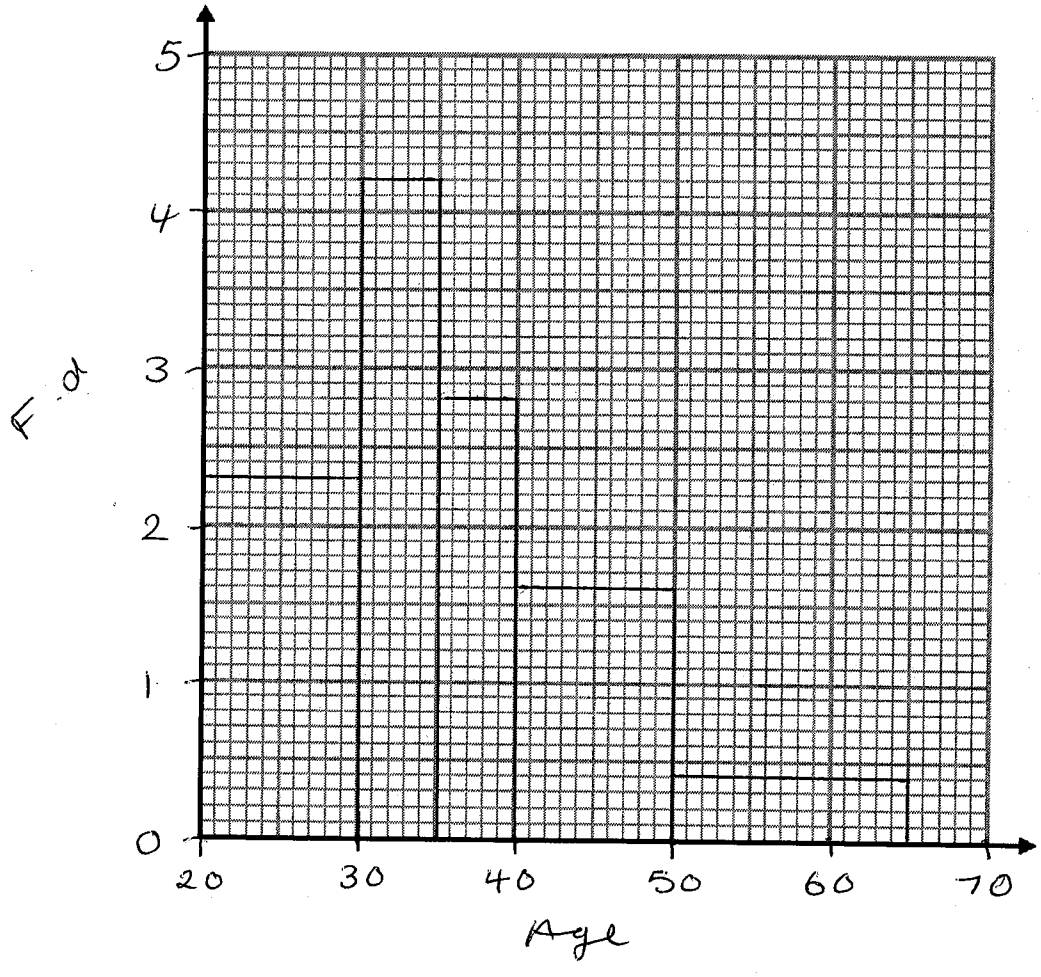
1 The table shows information about the age of 80 teachers.

Age (years)	Frequency	F.d
$20 < a \leqslant 30$	23	2.3
$30 < a \leqslant 35$	21	4.2
$35 < a \leqslant 40$	14	2.8
$40 < a \leqslant 50$	16	1.6
$50 < a \leqslant 65$	6	0.4

On the grid, draw a histogram for the information in the table.



(Total for question 1 is 3 marks)

2 Simplify fully $\dfrac{2x+4}{5x-15} \div \dfrac{x^2-8x-20}{2x^2-x-15}$

$$\frac{2x+4}{5x-15} \times \frac{2x^2-x-15}{x^2-8x-20}$$

$$\frac{2\cancel{(x+2)}(2x+5)\cancel{(x-3)}}{5\cancel{(x-3)}(x-10)\cancel{(x+2)}}$$

$$\frac{2(2x+5)}{5(x-10)}$$

$\dfrac{2(2x+5)}{5(x-10)}$

(Total for Question 2 is 3 marks)

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3 Prove that the sum of 3 consecutive even numbers is always a multiple of 6.

$2n + 2n + 2 + 2n + 4$

$6n + 6$

$6(n+1)$

(Total for Question 3 is 2 marks)

4 $V = IR$

$I = 4.29$ correct to 2 decimal places
$R = 16.173$ correct to 3 decimal places

Work out the upper bound for V.
Give your answer to 2 decimal places.

upper V = upper I × upper R

= 4.295 × 16.1735

= 69.47

69.47

(Total for Question 4 is 3 marks)

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5 (a) Write $3x^2 - 12x + 19$ in the form $a(x + b)^2 + c$ where a, b, and c are integers.

$3(x^2 - 4x) + 19$

$3((x-2)^2 - 4) + 19$

$3(x-2)^2 - 12 + 19$

$3(x-2)^2 + 7$

$3(x-2)^2 + 7$

(3)

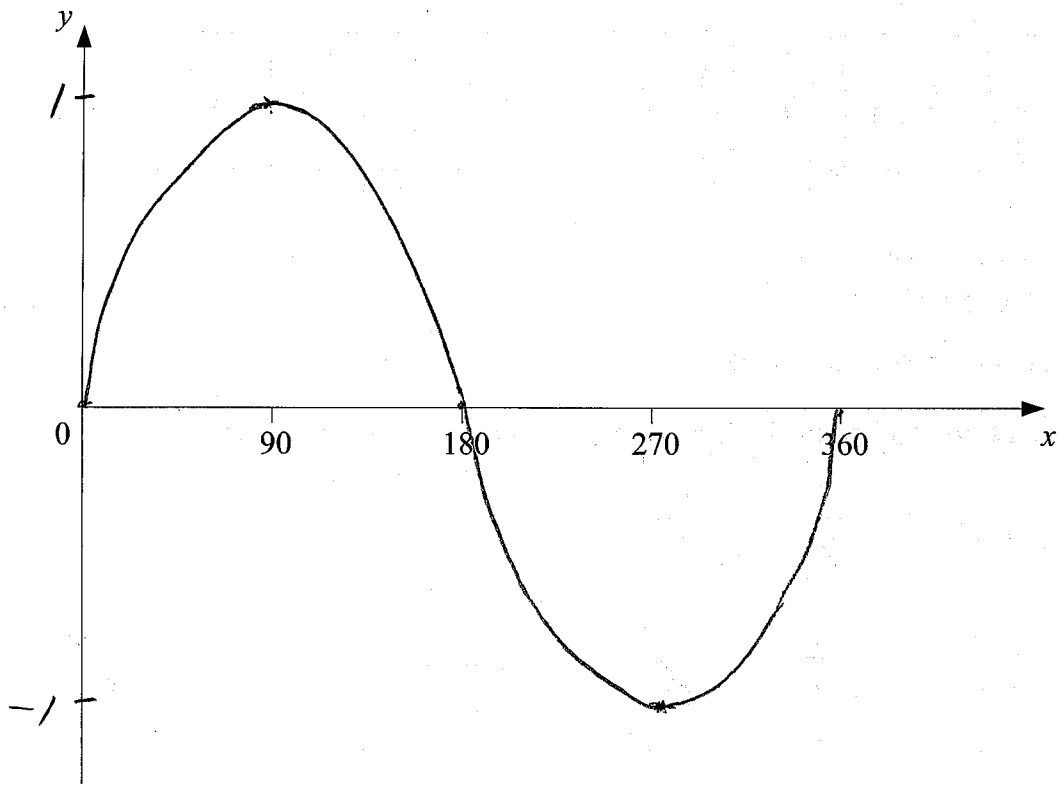
(b) Hence, or otherwise, write down the coordinates of the turning point of the graph of $y = 3x^2 - 12x + 19$

(2, 7)

(1)

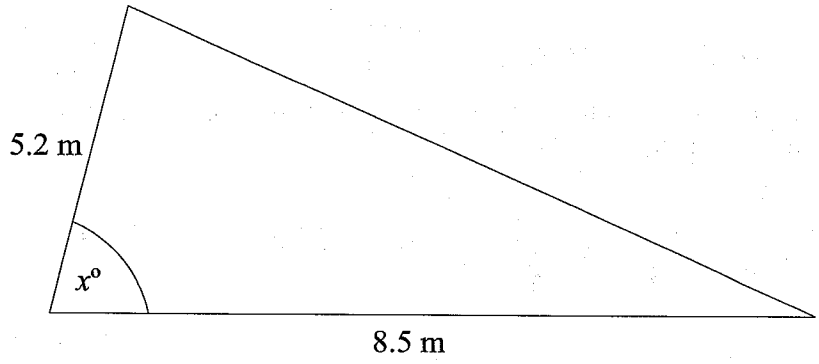
(Total for Question 5 is 4 marks)

6 Sketch the graph of $y = \sin x°$ for $0 \leq x \leq 360$



(Total for Question 6 is 2 marks)

7



The area of the triangle is 22 m^2
Work out the value of x.
Give your answer to 3 significant figures.

$$\frac{1}{2}(5.2)(8.5)\sin x = 22$$

$$\sin x = \frac{22}{\frac{1}{2}(5.2)(8.5)}$$

$$x = \sin^{-1}\left(\frac{220}{221}\right)$$

$$= 84.5$$

84.5 °

(Total for Question 7 is 4 marks)

8 Here are seven number cards.

1	1	1	2	2	3	3

Helen takes a card at random.
She does not replace the card.

Helen then takes another card at random.

(a) Calculate the probability that both cards have the same number on them.

$P(1,1) = \frac{3}{7} \times \frac{2}{6} = \frac{6}{42}$

$P(2,2) = \frac{2}{7} \times \frac{1}{6} = \frac{2}{42}$

$P(3,3) = \frac{2}{7} \times \frac{1}{6} = \frac{2}{42}$

$\frac{6}{42} + \frac{2}{42} + \frac{2}{42}$

$\frac{10}{42}$

(OR $\frac{5}{21}$) **(3)**

(b) Calculate the probability that the number on the first card Helen takes and the number on the second card Helen takes have a sum of 4.

$P(1,3) = \frac{3}{7} \times \frac{2}{6} = \frac{6}{42}$

$P(2,2) = \frac{2}{42}$

$P(3,1) = \frac{2}{7} \times \frac{3}{6} = \frac{6}{42}$

$\frac{6}{42} + \frac{2}{42} + \frac{6}{42} = \frac{14}{42}$

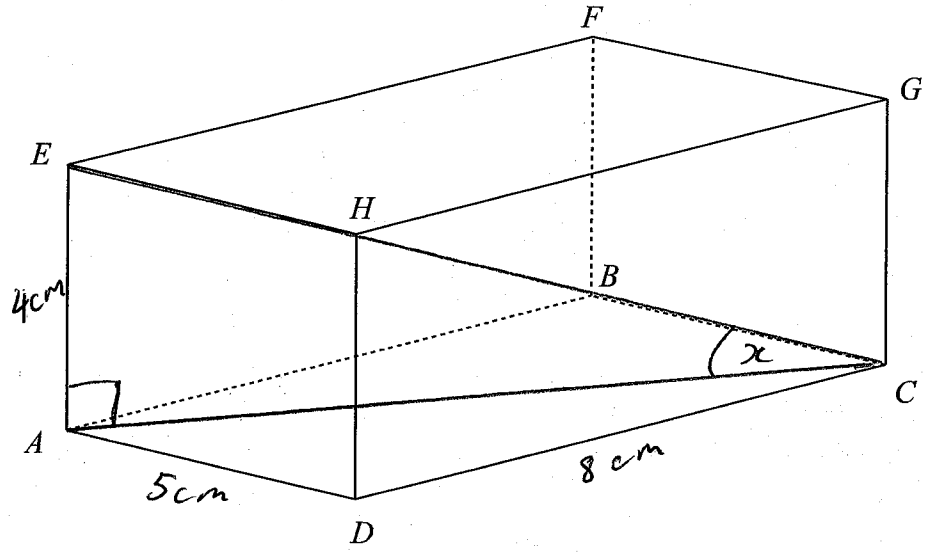
$\frac{14}{42}$

(OR $\frac{1}{3}$) **(3)**

(Total for Question 8 is 6 marks)

9 The diagram shows a cuboid *ABCDEFGH*.

$AE = 4$ cm
$AD = 5$ cm
$DC = 8$ cm



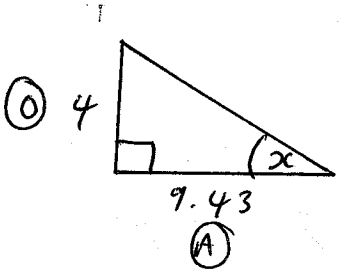
Calculate the size of angle *ECA*.
Give your answer correct to 3 significant figures.

$AD^2 + CD^2 = AC^2$

$5^2 + 8^2 = AC^2$

$AC = \sqrt{5^2 + 8^2}$

$= 9.43398...$



$\tan x = \frac{4}{9.43398}$

$x = \tan^{-1}\left(\frac{4}{9.43398}\right)$

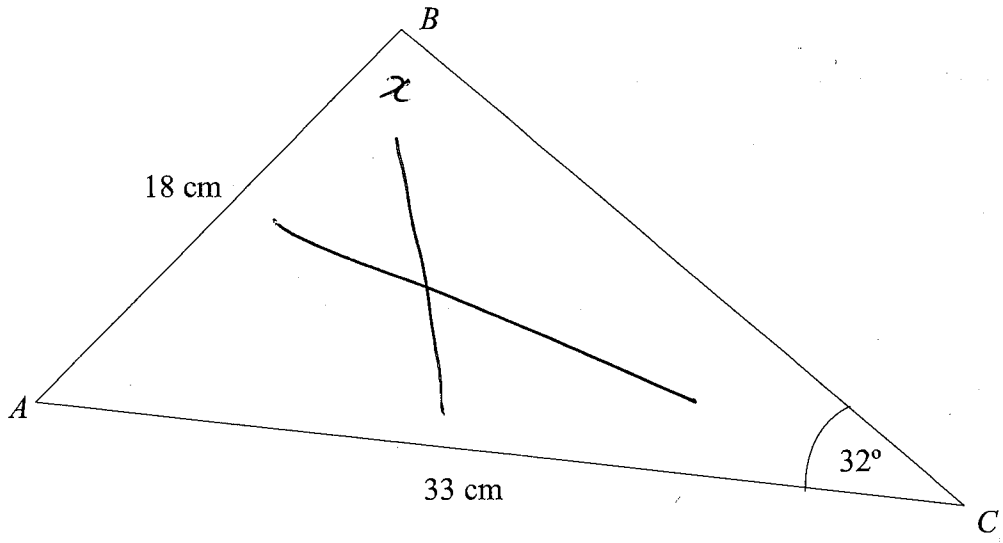
$= 22.9769$

$= 23.0$

23.0 °

(Total for Question 9 is 4 marks)

10



Angle *ABC* is obtuse.
Work out the size of angle *ABC*.
Give your answer to 3 significant figures.

$$\frac{\sin x}{33} = \frac{\sin 32}{18}$$

$$\sin x = \frac{\sin 32}{18} \times 33$$

$$= 0.97151\ldots$$

$$x = \sin^{-1}(\text{Ans})$$

$$= 76.29$$

x is obtuse $\therefore$ $x = 180 - 76.29$

$$= 103.707$$

$$= 104$$

104 °

(Total for Question 10 is 4 marks)